

Lab and Field Experiments for Ultrasonic, Radar, and LiDAR Sensors

This document contains the comprehensive details and results of all the practical experiments (laboratory, sea, and river trials) we conducted.

Lab Experiments

Measurement results of the Radar and ultrasonic sensors under several test conditions, compared against manual tape measurements. The trials were conducted at different locations inside the laboratory.

Ultrasonic Sensor - Lab Trials

The ultrasonic sensor has a wide beam width of 60°, making it highly affected by interference from nearby objects and surfaces, which required careful positioning to get accurate readings.

Test No.	Condition	Actual Distance (m)	Sensor Reading (m)	Observation
1	Sensor placed horizontally at the middle of the table facing the wall	3.83	1.33	Incorrect reading because the table surface interfered with the beam
2	Sensor placed vertically facing 5cm of water	0.52	0.52	
3	Sensor placed vertically facing 5cm of water from a farther distance	0.93	0.94	
4	Sensor placed horizontally at the table edge facing the wall	2.12	2.120	

Radar Sensor - Lab Trials

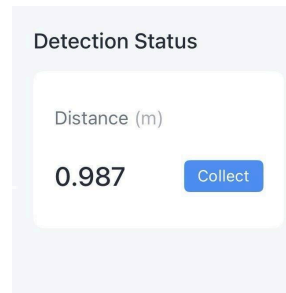
The radar sensor has a narrow beam width of 8°, allowing for more focused and accurate readings. However, due to this narrow beam, it is highly sensitive to tilt, as even a slight misalignment causes the signal to miss the target surface entirely.

Test No.	Condition	Actual Distance (m)	Sensor Reading (m)	Observation
1	Sensor placed horizontally at the edge of a table facing the wall	0.66	0.661	
3	Sensor placed horizontally at the	0.88	0.882	accurate reading despite the small table interference

	middle of the table facing the wall			because the Radar beamwidth is much narrower than the ULD
4	Sensor placed horizontally facing the wall from a longer distance	3.90	3.899	
5	Sensor placed vertically facing a water container in the lab	0.79	0.789	Waves are successfully reflecting from the water surface
6	Sensor placed horizontally facing a long hallway	12.5	4.285	Incorrect measurements because the sidewalls interfered with the beam



(a) Actual distance measured with tape



(b) Sensor reading (screenshot)

Figure 2: Test No. - Sensor placed vertically facing the floor

Lidar Sensor - Lab Trials

The LiDAR sensor has a narrow beam width of 0.5° , allowing for highly focused readings. In initial trials measuring solid surfaces like the ground and walls without a capacitor, the measurement error was approximately ± 10 cm, but after installing it, the accuracy of the readings improved. However, when tested over calm, clear water, the LiDAR sensor behaved differently. Unlike the radar sensor, it did not reflect off the water surface; instead, it penetrated the water and reflected off the bottom of the container. This resulted in incorrect readings because of the different travel speeds of the laser in air and water.

Test No.	Condition	Actual Distance (cm)	Sensor Reading (cm)	Observation
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1	Sensor placed vertically on the table, facing the floor	75	73	
2	Sensor placed vertically at a higher position, facing the floor	200	202	
3	Sensor placed vertically facing the floor, from a farther distance	445	449	
4	Sensor placed horizontally, facing the wall	100	108	large deviation from actual distance
5	Sensor placed horizontally facing 5 cm of water in a container	70	83	Incorrect measurement because the sensor did not reflect properly off the water surface

Sea Experiments

To evaluate sensor performance a field experiment was done in downtown Trieste Grignano seashores.



(a) Calm sea water observed during the trial

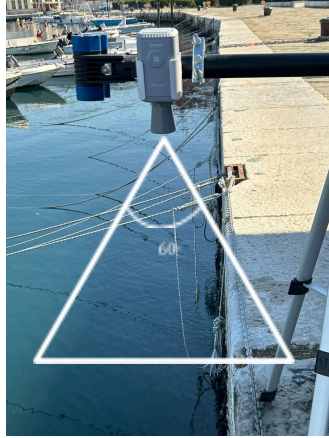


(b) Overview of the test setup

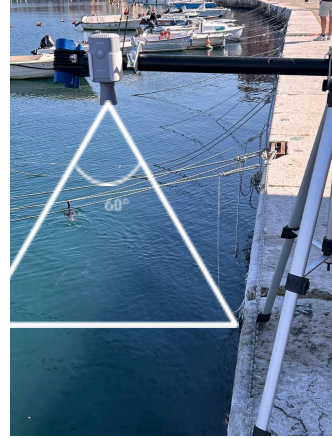
Figure 4: sensors test at the sea

Ultrasonic Sensor - Sea Trial

Sea trials revealed that the ultrasonic sensor's wide 60 acoustic beam requires careful installation. As shown in Figure 4(a), mounting the sensor too close to the pier caused the beam to intersect the concrete wall before reaching the water. This structural interference resulted in a false echo and an incorrect short reading (0.905 m). When the sensor was repositioned further outward Figure 4(b), the measurements were accurate (2.266 m). Therefore, mounting brackets must be designed long enough to prevent the sensor's beam from hitting adjacent structures.



(a) Sensor placed close to the pier edge



(b) Sensor placed away from the pier edge

Figure 4: Beamwidth Effect

Test No.	Condition	Actual Distance (m)	Sensor Reading (m)	Observation
1	Sensor placed vertically close to the pier edge, facing calm sea water (see Figure 2)	2.25	0.905	Incorrect measurements because the pier edge interfered with the beam
2	Sensor placed vertically away from the pier edge, facing calm sea water (see Figure 3)	2.25	2.266	accurate reading once the sensor was clear of any interference

Trial	Sensor Reading (m)	Ground Truth (m)	Error (m)
1	2.191	2.20	-0.009
2	2.196	2.20	-0.004
3	2.196	2.20	-0.004
4	2.196	2.20	-0.004
5	2.214	2.20	+0.014
6	2.214	2.20	+0.014
7	2.210	2.20	+0.010
8	2.210	2.20	+0.010
9	2.218	2.20	+0.018
10	2.218	2.20	+0.018
Average	2.206	2.20	
Standard Deviation	0.010		

Radar Sensor - Sea Trial

Test No.	Condition	Actual Distance (m)	Sensor Reading (m)	Observation
1	Sensor placed vertically facing the floor	0.96	0.964	
2	Sensor placed close to the pier edge facing calm sea water (see Figure 7)	2.48	2.468	accurate reading because the Radar beamwidth is much narrower than the ULD

Trial	Sensor Reading (m)	Ground Truth (m)	Error (m)
1	2.517	2.48	+0.037
2	2.468	2.48	-0.012
3	2.509	2.48	+0.029
4	2.514	2.48	+0.034
5	2.473	2.48	-0.007
6	2.507	2.48	+0.027
7	2.487	2.48	+0.007
8	2.484	2.48	+0.004
9	2.477	2.48	-0.003
10	2.476	2.48	-0.004
Average	2.491	2.48	
Standard Deviation	0.019		



Detection Status

Distance (m)

2.468

Collect

(a) Sensor mounted close to the pier edge

(b) Sensor reading (screenshot)

Figure 7: Test No. 2 - Radar sensor positioned close to the pier edge

River Experiments

To evaluate sensor performance a field experiment was done at the Timavo River located in the Province of Trieste in northeastern Italy. During the test, the air temperature was 29°C and the water temperature was 12°C.

The three sensors were mounted on a horizontal bar above the river (Figure 8). The Ultrasonic sensor was positioned first, since it has the widest beam and is therefore the most sensitive to placement, followed by the Radar sensor, and finally the LiDAR sensor. The bar was tilted to a series of fixed , and four consecutive readings were recorded from each sensor at every angle.



Figure 8: Sensor rig mounted above the Timavo River, Trieste

Because tilting the bar changes the physical height of each sensor above its mounting point, the expected (ground truth) reading at each angle was calculated geometrically

Since each sensor is mounted at a different point along the bar, each has its own arm length and therefore its own ground truth value at every angle: $L = 84$ cm for the Ultrasonic sensor, $L = 61$ cm for the Radar sensor, and $L = 40$ cm for the LiDAR sensor.

Angle (°)	Metric	Ultrasonic	Radar	LiDAR
0°	Ground Truth	2.620	2.620	2.620
	Reading 1 (m)	2.565	2.615	2.670
	Reading 2 (m)	2.568	2.616	2.660
	Reading 3 (m)	2.564	2.617	2.760
	Reading 4 (m)	2.566	2.618	2.680
	Mean (Average) (m)	2.566	2.617	2.692
	Standard Deviation (m)	0.0017	0.0013	0.0457
	Error (%)	-2.07%	-0.13%	+2.77%
10°	Ground Truth	2.766	2.726	2.689
	Reading 1 (m)	2.736	2.707	2.790
	Reading 2 (m)	2.725	2.702	2.760
	Reading 3 (m)	2.585	2.625	2.790
	Reading 4 (m)	2.584	2.624	2.802
	Mean (Average) (m)	2.658	2.664	2.785
	Standard Deviation (m)	0.0844	0.0462	0.0179
	Error (%)	-3.92%	-2.25%	+3.57%
15°	Ground Truth	2.837	2.778	2.724
	Reading 1 (m)	2.825	2.749	—
	Reading 2 (m)	2.823	2.761	—
	Reading 3 (m)	2.826	2.745	—
	Reading 4 (m)	—	2.772	—
	Mean (Average) (m)	2.825	2.757	—
	Standard Deviation (m)	0.0015	0.0122	—
	Error (%)	-0.45%	-0.76%	—
20°	Ground Truth	2.907	2.829	2.757
	Reading 1 (m)	2.888	0.956	2.680
	Reading 2 (m)	2.884	0.927	—
	Reading 3 (m)	2.885	0.958	—
	Reading 4 (m)	2.886	0.948	—
	Mean (Average) (m)	2.886	0.947	2.680
	Standard Deviation (m)	0.0017	0.0142	—
	Error (%)	-0.74%	-66.51%	-2.79%
25°	Ground Truth	2.975	2.878	2.789
	Reading 1 (m)	2.994	0.992	2.740
	Reading 2 (m)	2.974	0.998	—
	Reading 3 (m)	2.964	0.985	—
	Reading 4 (m)	2.965	0.964	—
	Mean (Average) (m)	2.974	0.985	2.740
	Standard Deviation (m)	0.0139	0.0148	—
	Error (%)	-0.03%	-65.78%	-1.76%
30°	Ground Truth	3.040	2.925	2.820
	Reading 1 (m)	3.038	—	—
	Reading 2 (m)	3.031	—	—
	Reading 3 (m)	3.029	—	—
	Reading 4 (m)	3.034	—	—
	Mean (Average) (m)	3.033	—	—
	Standard Deviation (m)	0.0039	—	—
	Error (%)	-0.23%	—	—
32°	Ground Truth	3.065	2.943	2.832

Angle (θ)	Metric	Ultrasonic	Radar	LiDAR
	Reading 1 (m)	—	—	3.640
	Reading 2 (m)	—	—	3.680
	Reading 3 (m)	—	—	3.710
	Reading 4 (m)	—	—	3.680
	Mean (Average) (m)	—	—	3.678
	Standard Deviation (m)	—	—	0.0287
	Error (%)	—	—	+29.86%

River Experiments — another Trial

Another field experiment was done at the Timavo River

Angle (θ)	Metric	Ultrasonic	Radar	LiDAR
0°	Ground Truth	2.620	2.620	2.620
	Reading 1 (m)	2.580	2.731	2.660
	Reading 2 (m)	2.575	2.590	2.720
	Reading 3 (m)	2.561	2.588	2.650
	Reading 4 (m)	2.621	2.570	2.701
	Mean (Average) (m)	2.572	2.620	2.677
	Standard Deviation (m)	0.0098	0.0747	0.0379
	Error (%)	-1.83%	-0.01%	+2.16%
11°	Ground Truth	2.805	2.765	2.687
	Reading 1 (m)	2.799	2.727	—
	Reading 2 (m)	2.788	2.735	—
	Reading 3 (m)	—	2.732	—
	Reading 4 (m)	—	2.727	—
	Mean (Average) (m)	2.793	2.730	—
	Standard Deviation (m)	0.0078	0.0039	—
	Error (%)	-0.41%	-1.26%	—
12°	Ground Truth	2.822	2.778	2.693
	Reading 1 (m)	—	2.728	—
	Reading 2 (m)	—	2.733	—
	Reading 3 (m)	—	2.724	—
	Reading 4 (m)	—	—	—
	Mean (Average) (m)	—	2.728	—
	Standard Deviation (m)	—	0.0045	—
	Error (%)	—	-1.79%	—
15°	Ground Truth	2.871	2.817	2.711
	Reading 1 (m)	2.848	—	—

Angle (θ)	Metric	Ultrasonic	Radar	LiDAR
	Reading 2 (m)	—	—	—
	Reading 3 (m)	—	—	—
	Reading 4 (m)	—	—	—
	Mean (Average) (m)	2.848	—	—
	Standard Deviation (m)	—	—	—
	Error (%)	-0.80%	—	—
17°	Ground Truth	2.904	2.842	2.722
	Reading 1 (m)	2.860	—	—
	Reading 2 (m)	2.860	—	—
	Reading 3 (m)	—	—	—
	Reading 4 (m)	—	—	—
	Mean (Average) (m)	2.860	—	—
	Standard Deviation (m)	0.0000	—	—
	Error (%)	-1.50%	—	—
21°	Ground Truth	2.968	2.892	2.745
	Reading 1 (m)	2.950	—	—
	Reading 2 (m)	2.952	—	—
	Reading 3 (m)	2.948	—	—
	Reading 4 (m)	2.956	—	—
	Mean (Average) (m)	2.952	—	—
	Standard Deviation (m)	0.0034	—	—
	Error (%)	-0.54%	—	—
23°	Ground Truth	2.999	2.917	2.757
	Reading 1 (m)	2.980	—	—
	Reading 2 (m)	—	—	—
	Reading 3 (m)	—	—	—
	Reading 4 (m)	—	—	—
	Mean (Average) (m)	2.980	—	—
	Standard Deviation (m)	—	—	—
	Error (%)	-0.63%	—	—
30°	Ground Truth	3.105	3.000	2.795
	Reading 1 (m)	3.093	—	—
	Reading 2 (m)	3.093	—	—
	Reading 3 (m)	3.083	—	—
	Reading 4 (m)	3.088	—	—
	Mean (Average) (m)	3.089	—	—
	Standard Deviation (m)	0.0048	—	—
	Error (%)	-0.51%	—	—
31°	Ground Truth	3.120	3.011	2.800
	Reading 1 (m)	—	—	—
	Reading 2 (m)	—	—	—
	Reading 3 (m)	—	—	—
	Reading 4 (m)	—	—	—
	Mean (Average) (m)	—	—	—
	Standard Deviation (m)	—	—	—
	Error (%)	—	—	—
32°	Ground Truth	3.134	3.023	2.805
	Reading 1 (m)	—	—	—
	Reading 2 (m)	—	—	—

Angle (θ)	Metric	Ultrasonic	Radar	LiDAR
	Reading 3 (m)	—	—	—
	Reading 4 (m)	—	—	—
	Mean (Average) (m)	—	—	—
	Standard Deviation (m)	—	—	—
	Error (%)	—	—	—
35°	Ground Truth	3.176	3.056	2.821
	Reading 1 (m)	—	—	—
	Reading 2 (m)	—	—	—
	Reading 3 (m)	—	—	—
	Reading 4 (m)	—	—	—
	Mean (Average) (m)	—	—	—
	Standard Deviation (m)	—	—	—
	Error (%)	—	—	—
41°	Ground Truth	3.256	3.119	2.850
	Reading 1 (m)	—	—	—
	Reading 2 (m)	—	—	—
	Reading 3 (m)	—	—	—
	Reading 4 (m)	—	—	—
	Mean (Average) (m)	—	—	—
	Standard Deviation (m)	—	—	—
	Error (%)	—	—	—